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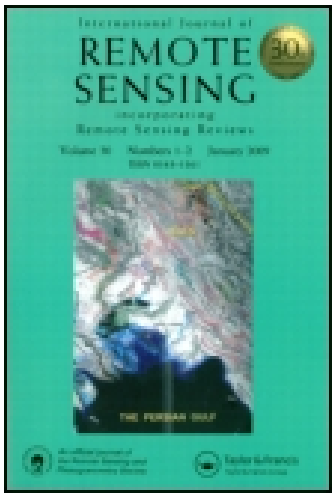

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Colour composite comparisons for agricultural assessments

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Colour composite comparisons for agricultural assessments

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Abstract. There have been extensive evaluations of different remote sensing digital classification techniques. In contrast, there has been very little systematic evaluation of various enhancements, such as different colour composites, for visual interpretations. This study examined three Landsat Thematic Mapper (TM) colour composites for visual classification of agricultural crops in an Imperial Valley, California study area. Accuracy results varied considerably with the composite from the best three bands as determined by transformed divergence calculations being more useful than an image obtained via principle components or the standard infrared simulated false colour composite.

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1. Introduction

There has been significant research on the comparison of digital classifications using different spaceborne remotely-sensed data for the same area and the comparison of different computer-based classification strategies with the same data. For example, there have been numerous efforts to evaluate the ability of Landsat Multispectral Scanner (MSS) data to digitally delineate agricultural and associated land covers. Several of those efforts have had difficulty. The MSS 80-m pixel size is occasionally too coarse for these environments and, frequently, the four available MSS visible and near-infrared spectral bands have not been adequate for signature differentiation. The inclusion of the Thematic Mapper (TM) sensor with improved radiometric sensitivity, 30-m pixel size, and seven spectral bands in the visible, near-infrared, middle-infrared and thermal-infrared on Landsats-4 and -5 generally provided more appropriate data for agricultural land-cover analysis (Crist 1984, DeGloria 1984, Pitts 1984). There have also been comparative examinations of the MSS and TM data for informational content (Anuta *et al.* 1984) and classification accuracy (Williams *et al.* 1984). The usefulness of higher spatial but lower spectral resolution SPOT data and more recently spaceborne radar has also been extensively examined.

A very important aspect of the appropriate use of remotely-sensed data for which little literature exists is the selection of the best image for visual interpretation. For many remote sensing applications, the most appropriate method of analysis may be visual interpretation. There is considerable concern among application scientists in remote sensing that analysis and education has become too digital. There is a perception that many scientists regularly use automated classification techniques for many applications which could be accomplished more accurately and at less expenditure of labour and resources via manual interpretation. Many experienced users of remote sensing argue that automated classification methods should only be accomplished after visual techniques have been fully assessed. The appropriateness

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of visual interpretation extends simply beyond the best accuracy of classification to costs, equipment and staffing requirements, available infrastructure, and repeatability (Philipson and Hafker 1980).

Visual interpretation of remotely-sensed data can have numerous advantages. It can be more accurate because it incorporates association information of surrounding pixels, such as texture and context. This can be easily accomplished by the human eye while digital classification has been generally limited to individual pixel analysis. This is changing as digital analysis attempts to duplicate visual methods by use of artificial intelligence, expert systems, and measures of context such as texture in classification. Nonetheless, visual interpretation will continue to be an important component of remote sensing applications. Visual interpretation of remotely-sensed data also requires less equipment and staffing resources. The equipment requirements are minimal only consisting of an image and perhaps overlay material and simple magnification.

Visual interpretation techniques are often quite familiar to members of the application community who have previously worked with aerial photography. One of the great differences between the interpretation of aerial photography and other remotely-sensed data is the range of enhancements possible with multiband imagery. Image selection is a very complex process and often data dependent. Image selection includes at the simplest level band selection and colour assignment for a tradition three-band colour composite. Increasingly, it also includes methods of data compression, such as ratioing and principle components, to allow information from more than three bands to be displayed, a vast range of spatial operators such as high and low pass filters and edge enhancements, and arithmetic manipulations of the data such as normalized green vegetation indexes.

As the ability of the remote sensing community to generate more data increases, the variety of enhancements also increases. Sheffield (1985) noted that the transition from four-band Landsat-MSS data to seven-band TM data increased the number of three band combinations to 35 and since any band can be assigned any colour, the number of different possible colour combinations is 210. Add to this the range of spatial operators and the variations in specific colour intensity for a given image, the number of enhancements from a given data set are infinite. The availability of hyperspectral and multi-sensor data greatly adds to the complexity of this issue.

A significant weakness in remote sensing is information on the selection of enhancements for various applications. There is very little literature on this topic. The geological community has the most experience on image enhancements but even for this application, the published literature is limited. Given the continued and perhaps increased importance of visual analysis, more direction should be provided on image selection. The purpose of this study was to compare the visual interpretation classification accuracy of agricultural crops using three different false colour composites (FCC) from the same TM data. These three images are the standard composite, a composite based on optimum band selection calculations, and an image representative of data compression.

2. Study area and data acquisition

Figure 1 is the standard FCC of the data used in this analysis. Bands 2, 3, and 4 are displayed as blue, green, and red respectively, to simulate colour infrared film. The TM data were collected by Landsat-4 on 12 December 1982. The band numbers and wavelengths for the seven TM bands are listed in table 1. TM pixels have a 30 m

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Figure 1. 12 December 1982 standard Landsat Thematic Mapper false colour composite for the Imperial Valley, California study area. Approximate scale 1 : 350 000.

spatial resolution—except the thermal band (band 6) which has 120 m resolution. TM data were collected with a dynamic range of 256 radiometric levels.

This scene is of the Imperial Valley, California, from the Salton Sea in the north-west to below the Mexican border. The border is very evident by the change in field size and shapes in the two countries. The primary land covers in this region are

Table 1. Thematic Mapper bands and subscene statistics

Band	Wavelength (μm)	Energy type	Mean value	Standard deviation
1	0·45– 0·52	Reflected blue	59	4·0
2	0·52– 0·60	Reflected green	25	2·5
3	0·63– 0·69	Reflected red	25	5·9
4	0·76– 0·90	Reflected near-IR	49	18·9
5	1·55– 1·75	Reflected mid-IR	39	7·9
6	10·40–12·50	Emitted thermal	106	1·6
7	2·08– 2·35	Reflected mid-IR	16	5·9

agricultural fields, mostly irrigated, in the image centre and desert landscapes including some bare sands and xerophytic vegetated areas to the east and west. Water surfaces include the portion of the Salton Sea, small playas, the New and Alamo Rivers passing through the agricultural region, and various irrigation canals such as the All American and East Highland. Rock outcrops of the Chocolate Mountains are in the northeastern corner. Figure 1 is an approximately 4000 square kilometer portion of the original Landsat scene.

The warm climate, relatively fertile soils, and availability of water for irrigation from the Colorado River allows a wide range of crops. Among the more prominent crops are lettuce, sugar beets, and cotton. Feed lot beef cattle are also found here. The continuous growing season allows for up to seven cuttings of alfalfa per year. Scattered within the agricultural area are a number of towns including Brawley, El Centro, Calexico, and Mexicali.

For this study, a subset of agricultural fields was selected. This subset consisted of 124 contiguous fields in the north-east of the agricultural area. These fields totaled 80 422 30 m pixels, and 5980 ha. Field cover information is collected monthly by the Imperial Valley Irrigation Board. During December 1982, the reported crop types for this data subset included: beets, cotton, alfalfa, fallow, onions, and broccoli. The data subset crops are identified in figure 2. Table 1 contains information on the spectral values in each band for this subscene. The thermal band has a typically very low standard deviation. Band 4, near-infrared, has a very high standard deviation as a result of the contrast between highly reflecting vigorous green vegetation and low reflecting fallow fields and especially wet soils.

There were several small fields identified as feedlots and leaching which were eliminated from the data set. In addition, the alfalfa fields were initially identified as both new and old but were combined in this analysis. The range of spectral signatures for the alfalfa fields reflect the rapid changes in this crop due to frequent cutting. The generally large field size in this area, averaging over 40 ha, is ideal for classification using spaceborne data.

3. Methodology

Three different FCCs were displayed on a colour monitor for visual interpretation. For each display, ten interpreters were given the crop types for 14 fields and asked to identify the crops in the remaining 110 fields. No interpreter classified more than one display. The accuracy of these interpretations were assessed using the crop information provided by the Imperial Valley Irrigation Board. The three images

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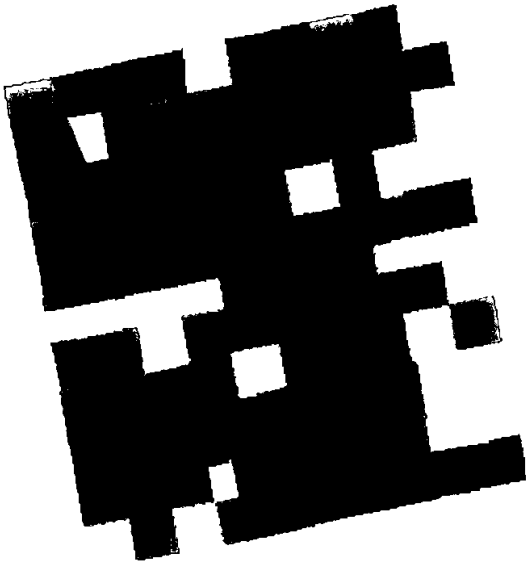

Figure 2. The field boundaries and crop types for the specific study area. Approximate scale 1 : 80 000. Class names: red, beets; pale blue, cotton; green, alfalfa; brown, fallow; yellow, onions; dark blue, broccoli.

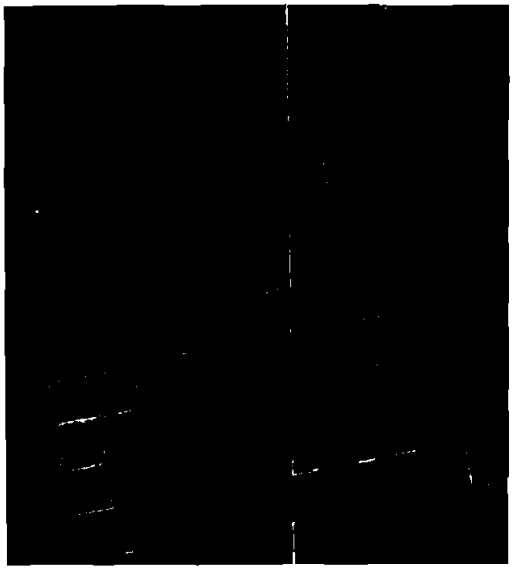

Figure 3. Standard false colour composite for field identifications. Bands 2, 3, and 4 in blue, green, and red. Approximate scale 1 : 80 000.

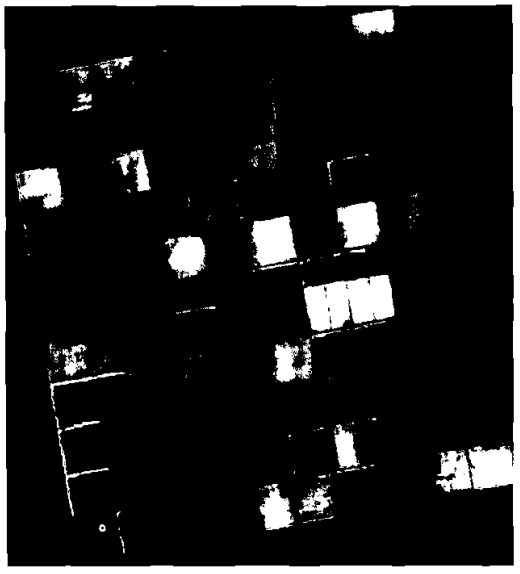

Figure 4. Image using optimal bands based on transformed divergence calculations. Bands 3, 5, and 4 in blue, green, and red. Approximate scale 1 : 80 000.

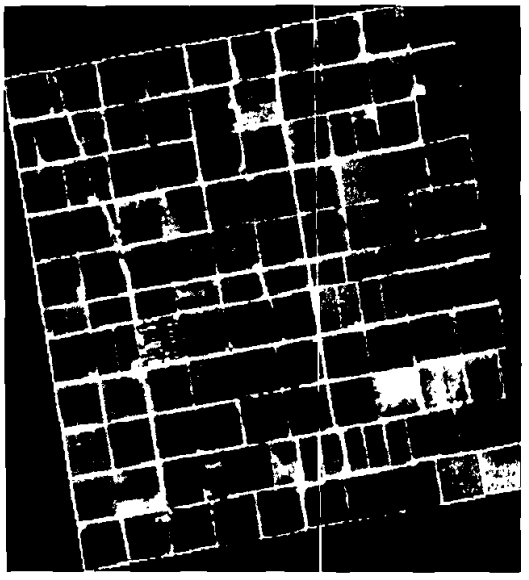

Figure 5. False colour composite from principal component images 1, 3, and 2 in blue, green and red. Approximate scale 1 : 80 000.

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were the standard TM false colour composite, a combination from the best three bands identified by transformed divergence, and the first three principle component data reduction images.

3.1. *Standard image composite*

The standard image for multispectral data is selected to simulate colour infrared photography because that is familiar to most users. With TM data this is bands 2 (visible green), 3 (visible red), and 4 (near-infrared) displayed in blue, green, and red respectively. Figure 3 is that image for this study. In that image, areas of high near-infrared reflectance, healthy green vegetation, are bright red tones. That is very evident for beets. The fallow fields areas are a blue-grey tone. Alfalfa is easily confused as it has a range of tones. The mature alfalfa is spectrally similar to the beets and the less mature alfalfa is not separable from the fallow fields. Cotton also does not have an unique tone on this image.

The standard FCC image tones are easily understood by most individuals but are not necessary the most useful bands. Most previous studies have indicated that an image created from different portions of the electromagnetic spectrum provides the most information. In the standard composite, two of the three bands are in the visible portion. Also, the mid-infrared bands of TM, especially band 5, have been shown very useful in many applications such as soils, geology, arid and semi-arid landscapes, and wetlands. There is not a mid-infrared band in the standard FCC. TM data, because it contains a visible blue band, can also be used to provide a normal colour image (1, 2, and 3). While familiar to all, this composite frequently does not provide maximum informational content because it uses three visible bands and also may suffer from haze and atmospheric interference in these shorter wavelengths.

3.2. *Optimal band selection*

There has been considerable development of band selection techniques from multi-band data. These techniques normally depend on a measure of statistical distance between the spectral signatures of selected features within a scene. The primary intent of these efforts has been data reduction to minimize the number of bands for digital analysis (Mausel *et al.* 1990). Data reduction for this purpose has become of less importance with most available, seven or fewer band, multi-spectral sensors as increased processing power, even in personal computers, allows analysis of large volume data sets. Band selection for data reduction in digital analysis may be of greater utility in hyperspectral data sets. Band selection continues to be useful for creation of images for visual interpretation. There have been a number of different band selection or separability techniques developed.

Chavez *et al.* (1982) suggested a measure called the Optimum Index Factor (OIF) for band selection from multiband data. The OIF is based on the variance and the correlation among the different bands. It weighs the variance of the individual bands by using their standard deviations and the correlations between the bands by using their correlation coefficients.

Dwivedi and Rao (1992) used an OIF to select the best TM three band image composition for delineating salt-affected soils of the Indo-Gangetic alluvial plain. They considered twenty possible images and determind that the band combination of TM 1, 3, and 5 contained the best information content based on OIF values. The

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standard 2, 3, and 4 image ranked relatively very low (twelfth). However, in comparing the OIF ranking to visual interpretation accuracy, there was not good correlation. The 1, 3, and 5 FCC did have the highest classification accuracy but the other band combinations did not show any consistency in correspondence between OIF values and mapping accuracy.

Sheffield (1985) used a different method for band selection with TM data. He suggested selecting those bands whose variance-covariance matrices produces an ellipsoid of maximum volume in statistical space. Using several different data sets, he concluded that bands 1 (visible blue), 4 (near-infrared), and 5 (mid-infrared) in blue, red, and green was the best FCC. In his study, the natural colour composite and the standard simulated infrared FCC were not very useful. In an independent study, a selected set of interpreters also found the 1, 4, and 5 combination very useful (Colvocoresses 1983). Kenk *et al.* (1988) also identified TM bands 1, 4, and 5 as the best image composition for visual discrimination of objects.

Among the other measures of spectral separability are divergence, transformed divergence, Bhattacharyya distance, and Jeffreys-Matusita distance. Mausel *et al.* (1990) compared these four separability measures for selecting the best four bands from eight multi-spectral video bands of an agricultural scene. That study found transformed divergence and the Jeffreys-Matusita distance had near perfect correlation with classification accuracy. Transformed divergence was selected as the band selection method for this study.

Transformed divergence, which is calculated from the means and covariance matrices of each spectral class or training site, is a measure of statistical distance between class or site pairs of interest and provides information on their separability. This separability is an indirect estimate of the likelihood of correct classification between groups of different band combinations (Swain *et al.* 1971). It is an estimate of information usually obtained by the time-consuming and expensive process of actual classification and accuracy evaluations. A discussion of transformed divergence, including some of its disadvantages, can be found in Swain *et al.* (1971), Swain and Davis (1978), and Latty and Hoffer (1980). Transformed divergence can be used to examine intraclass and interclass variability of land-covers and to determine the number and specific data bands most useful for classification, or in this study for creation of a FCC.

Transformed divergence calculations were employed to identify the best three bands for separating crop types in this Imperial Valley data (Haack *et al.* 1986). These calculations were based on the spectral signatures for twenty agricultural fields. The best three bands identified were 3 (visible red), 4 (near-infrared), and 5 (mid-infrared).

This agrees with the results of Dottavio and Dottavio (1984) who determined that TM-3, -4, and -5 were the best bands for vegetation classification. However, their study focus was for wetlands. Lauer and Waltz (1983) and Hass and Waltz (1983) preferred the use of TM bands 3, 4, and 5 over the standard 2, 3, and 4 infrared composition. In a subjective evaluation of TM band combinations for rangeland assessment, Haas (1992) suggests that the TM-3, -4, and -5 FCC is best for vegetation assessment but that roads and other bare ground features were better recognized from the TM-2, -4, and -7 combination.

A difficulty with these band selection techniques is that the optimal bands for statistical separability are not necessarily the best bands for visual interpretation (Mausel *et al.* 1990). An additional difficulty is that even after the best bands are

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selected, it is not clear what colour combinations to apply to the bands. There are six possible images which can be obtained by use of three colours and three given bands. Sheffield (1985) suggests that since the human eye is most sensitive to green, next to red, and least to blue, that when there is information on the variances of the selected bands, green be assigned to the band of greatest variance, then red, and finally blue. In this study, all six images were examined and the best image subjectively selected. That image is bands 3, 5, and 4 in blue, green, and red and is figure 4. On this image, the beets are quite distinct as red tones, the alfalfa is generally gold toned, and while the cotton and fallow areas are both blue-grey, the fallow fields are generally darker. The broccoli is unique but the onions, as expected for these row crops of limited biomass, are similar to the fallow areas.

3.3. *Principal components data reduction*

A common variation on band selection is data reduction by use of ratioing or more frequently principle components (Ingebritsen and Lyon 1985, Singh and Harrison 1985, Fung and LeDrew 1987). Statistically, the use of the first three principal component images in a false colour composite presents as much information as possible in three colours. For a Landsat-MSS image, these three components will often account for more than 99 per cent of scene variance (Lilisand and Kiefer 1979). A problem in principal component images is that colours of features are very data dependent and no common colour interpretation rules can be temporally or spatially extended.

A principal components algorithm was applied as a data reduction method to this TM data. From the resulting seven principal component bands, the first three were selected for a FCC. The colour combination was for principle component bands 1, 3, and 2 in blue, green, and red respectively (figure 5). This image was also selected subjectively from the six possible colour combinations. In this image there are no familiar red tones. The beets are a distinctive light blue and the onions a dark green. The fallow fields are generally a yellow-green but not greatly different than the cotton. The alfalfa is also not an unique signature.

The principal components image would be expected to be the most effective for visual interpretation because it contains information from all seven TM bands. A histogram equalization enhancement was applied to all images prior to interpretation to maximize the colour contrast.

4. Interpretation results and conclusions

The interpreters were asked to assign one of the six cover types to each of the 110 fields based on 14 known fields. The accuracies were measured by the percent of correctly identified fields. The results for the manual interpretations are found in table 2.

There are considerable differences in the accuracy of visual interpretations of the various TM images. The standard three-band false colour composite provided the least information, an average of 62 per cent correct fields. This compares to 76 per cent correct for the three principal components image and 84 per cent for the best three bands.

The classification accuracies for the optimal band and principle components images were acceptable despite the spectral similarity of some cover types. These

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Table 2. Correctly identified fields.

	Percent correct	Standard deviation
Standard FCC (2, 3, 4-B, G, R)	62·3	10·2
Transformed divergence (3, 5, 4-B, G, R)	84·2	7·1
Principal components (1, 3, 2-B, G, R)	75·8	5·7

results were comparable to previous examinations of TM classification for vegetables (Williams *et al.* 1986). However, the interest in these results is not their absolute accuracy but their relative differences.

It is not surprising that the standard FCC has the lowest results but the magnitude of the difference is greater than anticipated. The relatively low values for the principle components image is unexpected because in theory, it contains most of the spectral differences in all seven bands. The highest results from the transformed divergence selection is an important result. It confirms for these data, the value of using bands from different portions of the electromagnetic spectrum, the usefulness of the mid-infrared bands, and that computationally intense data reduction methods are not necessary.

There are many possible extensions of this study. It could be extended to a larger number of interpreters and other band and colour combinations. This study should also be repeated in other agricultural areas containing different crops and cropping practices. Most important, this study indicates that greater attention should be given to the issue of image selection for visual interpretations. The increasing requirement of geographical information systems for current, reliable spatial data is being resolved by remote sensing techniques. For many applications that information extraction is by visual interpretation. The quality of that information will be determined by the image composite thus requiring greater understanding of the image selection process.

References

ANUTA, P. E., BARTOLUCCI, L. A., DEAN, M. E., LOZANO, D. F., MALARET, E., MCGILLEM, C. D., VALDES, J. A., and VALENZUELA, C. R., 1984, Landsat 4 MSS and Thematic Mapper data quality and information content analysis. *I.E.E.E. Transactions on Geoscience and Remote Sensing*, **22**, 152–159.

CHAVEZ, P. S. JR., BERLIN, G. L., and SOWERS, L. B., 1982, Statistical methods for selecting Landsat-MSS ratios. *Journal of Applied Photogrammetric Engineering*, **8**, 23–30.

COLVOCORESSES, A. P., 1983, Presentation at the Symposium on Landsat-4 Scientific Characterization-Early Results. Greenbelt, Maryland (Washington, DC: NASA).

CRIST, E. P., 1984, Comparison of coincident Landsat 4 MSS and TM data over an agricultural region. *Proceedings of the American Society of Photogrammetry and Remote Sensing Annual Meeting* (Falls Church, VA: ASPRS), pp. 508–517.

DEGLORIA, S. D., 1984, Spectral variability of Landsat-4 Thematic Mapper and Multispectral Scanner data for selected crop and forest types. *I.E.E.E. Transactions on Geoscience and Remote Sensing*, **22**, 303–311.

DOTTAVIO, C. L., and DOTTAVIO, F. D., 1984, Potential benefits of new satellite sensors to wetland mapping. *Photogrammetric Engineering and Remote Sensing*, **30**, 599–606.

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DWIVIDI, R. S., and RAO, B. R. M., 1992, The selection of the best possible Landsat TM band combination for delineating salt-affected soils. *International Journal of Remote Sensing*, **13**, 2051-2058.

FUNG, T., and LEDREW, E., 1987, Application of principle components to change detection. *Photogrammetric Engineering and Remote Sensing*, **53**, 1649-1658.

HAACK, B., BRYANT, N., and ADAMS, S., 1986, Landsat Thematic Mapper digital information content for agricultural environments. *Proceedings of the American Society of Photogrammetry and Remote Sensing Annual Meeting* (Falls Church, VA: ASPRS) **5**, 260-268.

HAAS, R. H., 1992, Landsat Thematic Mapper products for rangeland assessment. *Geocarto International*, **7**, 27-33.

HAAS, R. H., and WALTZ, F. A., 1983, Evaluation of Thematic Mapper data for natural resources assessment. *Proceedings Eighth Pecora Symposium, Sioux Falls, South Dakota* (Washington, D.C.: United States Geologic Survey), pp. 122-133.

INGEBRITSEN, S. E., and LYON, R. J. P., 1985, Principle components analysis of multitemporal image pairs. *International Journal of Remote Sensing*, **6**, 687-696.

KENK, E., SONDHEIM, M., and YEE, B., 1988, Methods for improving accuracy of Thematic Mapper ground cover classifications. *Canadian Journal of Remote Sensing*, **14**, 17-31.

LATTY, R. S., and HOFFER, R. M., 1980, Waveband evaluation of proposed Thematic Mapper in forest cover classification. *Proceedings of the Fall Technical Meeting, ACSM-ASP, Niagara Falls, New York* (Falls Church, VA: ASPRS), pp. RS-2-D-1 to RS-2-D-12.

LAUER, D. T., and WALTZ, F. A., 1983, Landsat-4 investigation of Thematic Mapper and Multispectral Scanner applications. *Proceedings of the Symposium on Landsat-4 Scientific Characterization—Early Results* (Greenbelt, Maryland: NASA).

LILLESAND, T. M., and KIEFER, R. W., 1979, *Remote Sensing and Image Interpretation* (New York: John Wiley and Sons).

MAUSEL, P. W., KRAMBER, W. J., and LEE, J. K., 1990, Optimum band selection for supervised classification of multispectral data. *Photogrammetric Engineering and Remote Sensing*, **56**, 55-60.

PHILIPSON, W. R., and HAFKER, W. R., 1980, Manual versus digital Landsat analysis for classification of remotely sensed data. *Proceedings of the American Society of Photogrammetry Fall Meeting* (Falls Church, VA: ASPRS), pp. RS-3-D-1-10.

PITTS, D. E., BADHWAR, G. D., THOMPSON, D. R., HENERSON, K. E., SHEN, S. S., SORENSON, C. T., and CARNES, J. G., 1984, Evaluation of corn/soybeans separability using Thematic Mapper simulator data. *I.E.E.E. Transactions on Geoscience and Remote Sensing*, **22**, 312-317.

SHEFFIELD, C., 1985, Selecting band combinations from multispectral data. *Photogrammetric Engineering and Remote Sensing*, **51**, 681-687.

SINGH, A., HARRISON, A., 1985, Standardized principle components. *International Journal of Remote Sensing*, **6**, 883-896.

SWAIN, P. H., and DAVIS, S. M., editors, 1978, *Remote Sensing: The Quantitative Approach*, (New York: McGraw Hill).

SWAIN, P. H., ROBERTSON, T. V., and WACKER, A. G., 1971. *Comparison of the Divergence and B-distance in Feature Selection* (West Lafayette, Indiana: Laboratory for the Application of Remote Sensing, Purdue University), 12 pp.

WILLIAMS, D. L., IRONS, J. R., MARKHAM, B. L., NELSON, R. F., TOLL, D. L., LATTY, R. S., and STAUFFER, M. L., 1984, A statistical evaluation of the advantages of Landsat Thematic Mapper data in comparison to Multispectral Scanner data. *I.E.E.E. Transactions on Geoscience and Remote Sensing*, **11**, 294-301.

WILLIAMS, V. L., PHILIPSON, W. R., and PHILPOT, W. D., 1986, Vegetable crop inventory with Landsat TM data. *Proceedings of the American Society of Photogrammetry and Remote Sensing Annual Convention* (Falls Church, VA: ASPRS), pp. 290-296.

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